

Agenda:

- Basics of programming
- Fundamentals of python
- Variables, Data Types
Operators
- Console IO

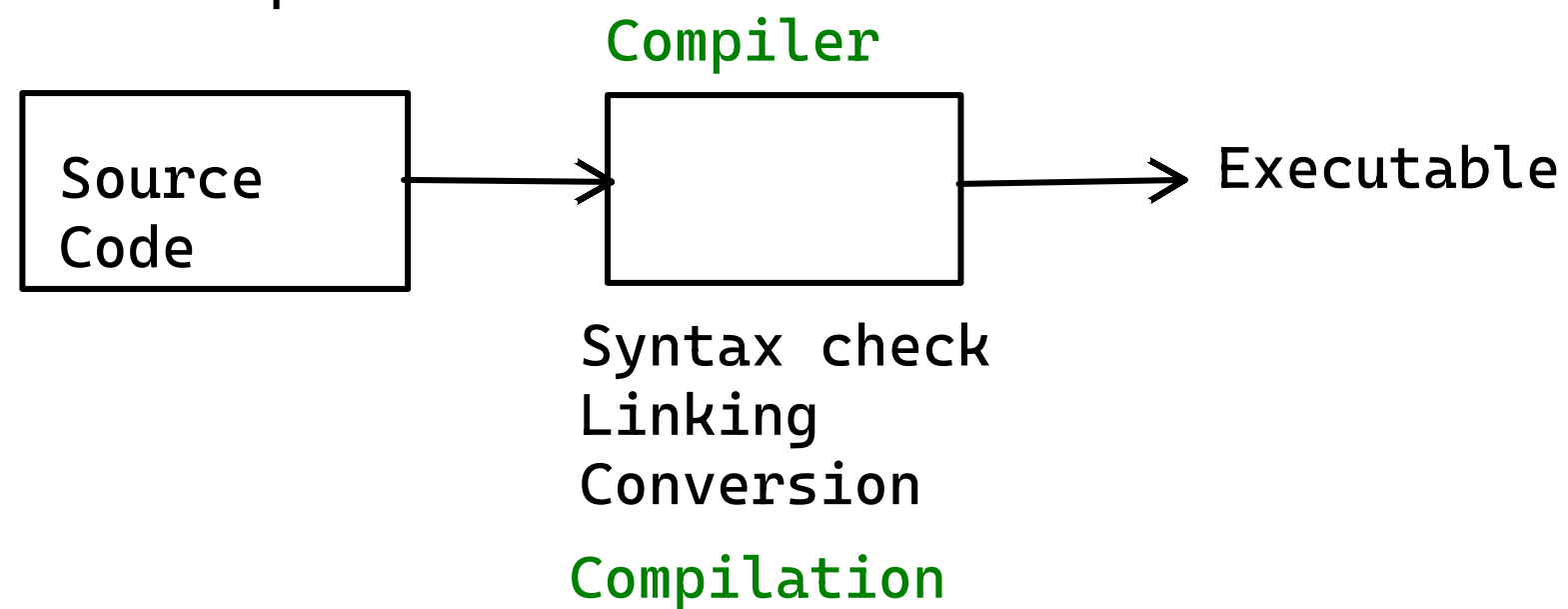
Timings:

9.30 to 11.15AM - P1
11.15 to 11.30 - break
11.30 to 1.15 -p2
1.15 to 2.15 - lunch break
2.15 to 3.45 - p3
3.45 to 4.00 - break
4.00 to 5.30 -p4

- 1/0 - Binary Number System
- High Level Language to Machine Level language
- Two:

1. Compilation
2. Interpretation

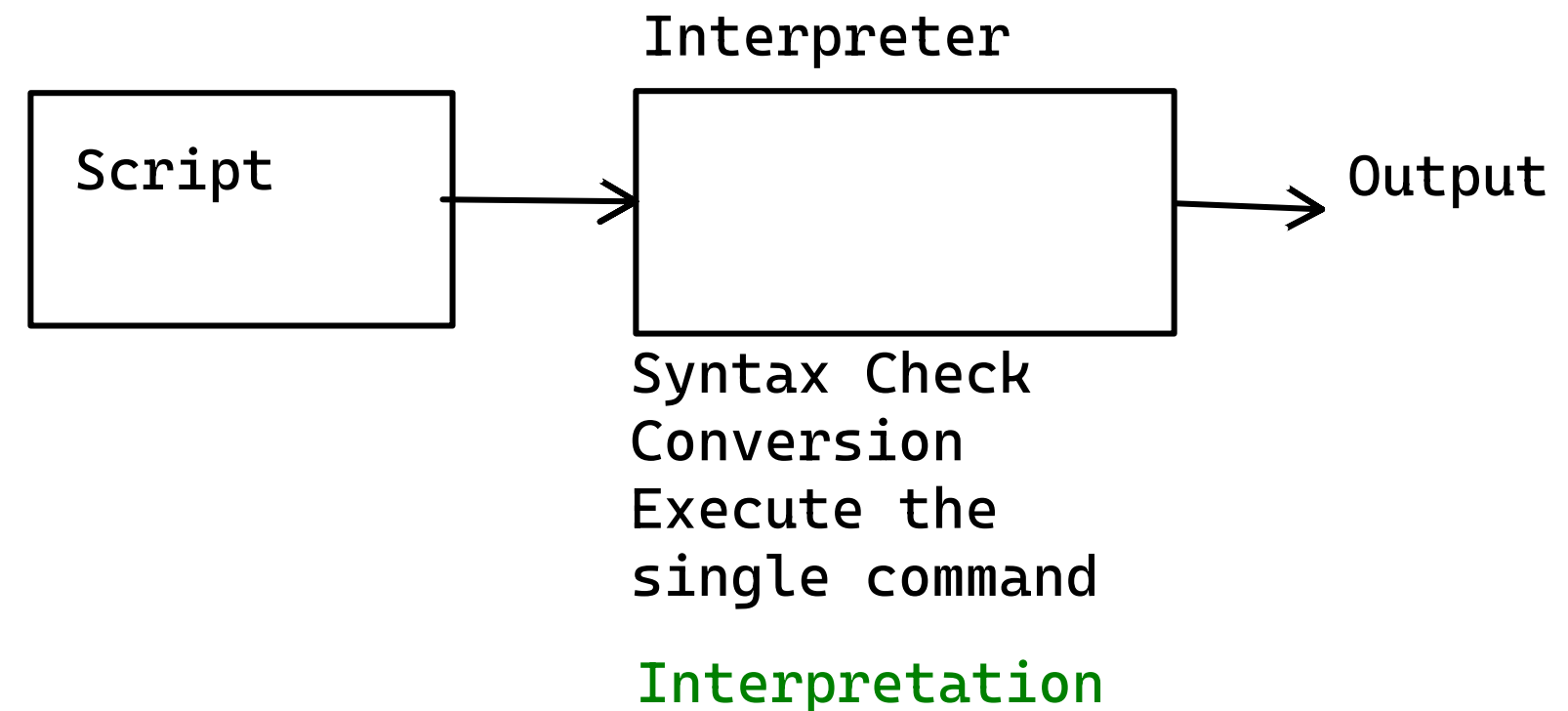
1. Compilation



C, C++, go, rust....

Pros & Cons

- No additional Software
- Secure
- Needs additional time for compilation
- granular control
- Expertise is needed
- Faster



Python, JavaScript....

- Needs Interpreter
- Not secure
- No need of additional time
- less granular control more abstraction
- less faster compare to comiled langauages

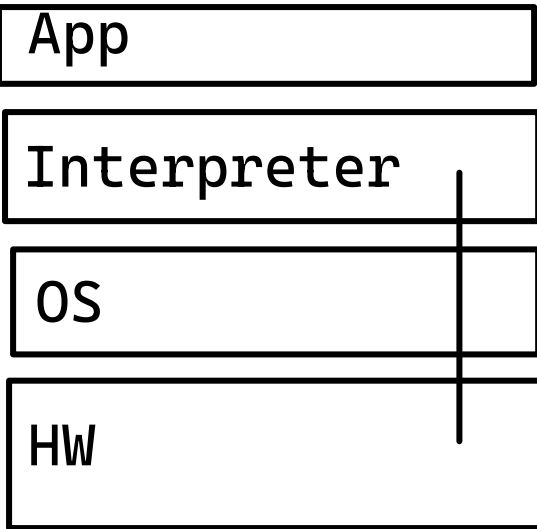
Python:

- Interpreted Language
- Open Source language
- Platform Independent language:App can be run on any combination of HW & OS without modification

Open Source:

Source Code is available - even for modification & Redistribution
GPL,MIT, APACHE LICENSE

WORA



Guido Van Rossum

Interpreter is Dependent on Architecture

WIN/LIN/MAC/ ...

... 128/64/32/16/8.....

- General purpose language: System Apps, Web Applications, CLIs, ML/AI, DS.....
- Dynamically Typed Language (later)

Versions of Python

2.x.x
3.x.x (latest)

2.x.x ~~3.x.x~~

3.9.15 3.12.12 3.15.3

Interpreters of Python

Cython - C+Python
Jython - Java + Python
RPython - R + Python
Pypi - python interpreter written in python

Lab: Installation of Python

How to learn a programming language

- How to store temporary data
- How to do conditions
- How to repeat
- What is the base paradigm

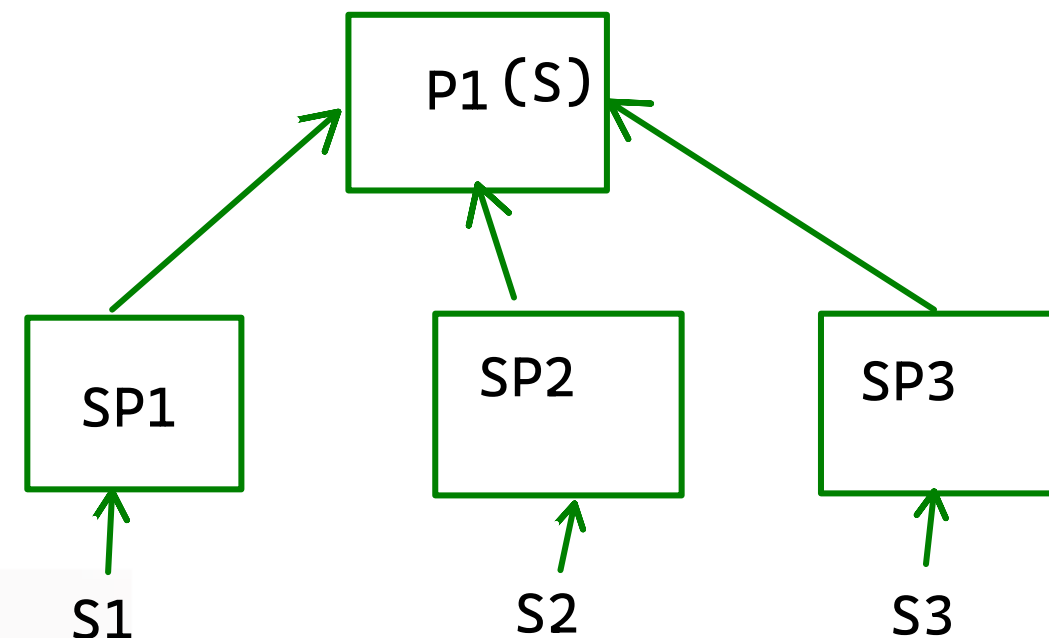
Base Paradigm of python

- Functional programming language

Other Paradigms:

- Object Oriented
- procedural

Functional Programming language



Example:

Scenario: You are a staff, You get CSVs or data from various dept
You have to do analysis, compile, provide a report for the
data received - HTML/PDF

Problem:

You are a staff, You get CSVs or data from various dept
You have to do analysis, compile, provide a report for the
data received – HTML/PDF

Sub Problems:

- | | |
|-------------------------|---|
| How to read the data | - read using python pandas library |
| Clean the data | - Missing data analysis, fill with default values |
| How to perform analysis | - Computations to be done |
| How to generate report | - Reportlab/fpdf.... tool to generate report |

Ways of interaction with Python

- Interactive Mode
 - Script Mode (.py)
 - Notebook Mode/Jupyter Notebook/ Google Colab (.ipynb)
- Data Analyst / Scientist

How to Store Data Temporarily

Variable: Place holder to data

Variable Identifier	Assignment operator	Actual Value
price	=	23.45

```
price
print(price)
```

Rules:

- alpha_numeric or _
- Cannot start with a number
- must not be a keyword

```
['False', 'None', 'True', 'and', 'as', 'assert', 'async', 'await', 'break', 'class', 'continue', 'def', 'del', 'elif', 'else', 'except', 'finally', 'for', 'from', 'global', 'if', 'import', 'in', 'is', 'lambda', 'nonlocal', 'not', 'or', 'pass', 'raise', 'return', 'try', 'while', 'with', 'yield']
```

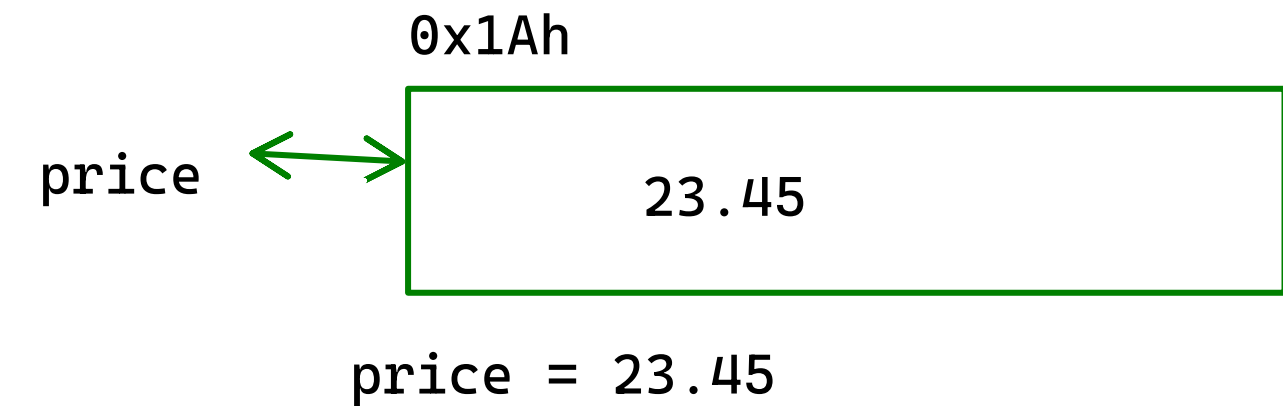
Conventions:

- Have meaningful names
- snake_case

Question: Identify Valid/ Invalid

- | | |
|------------|----------|
| 1. vishwas | 6. _123 |
| 2. var | 7. _abc |
| 3. else1 | 8. abc_! |
| 4. def | |
| 5. !123 | |

1. Valid
2. Valid
3. Valid
4. Invalid
5. Invalid
6. Valid
7. Valid
8. Invalid



```
>>> price = 23.45
>>> price
23.45
>>> print(price)
23.45
>>> type(price)
<class 'float'>
>>> age = 18
>>> age
18
>>> type(age)
<class 'int'>
>>> name = "Vishwas K Singh:
File "<python-input-7>", line 1
    name = "Vishwas K Singh:
          ^
SyntaxError: unterminated string literal (detected at line 1)
>>> name = "Vishwas K Singh"
>>> name
'Vishwas K Singh'
>>> type(name)
<class 'str'>
>>>
```

Data Types

- int
- float
- str
- bool
- complex

```
>>> v8 = 23.45
>>> type(v8)
<class 'float'>
>>> v9 = 2345e-100
>>> v9
2.345e-97
>>> v9 = 2345e-2
>>> v9
23.45
>>> v10 = 2345e-2
>>> v10
23.45
>>> v11 = .02345e2
>>> v11
2.345
>>> v11 = .2345e2
>>> v11
23.45
>>>
```

```
>>> v1 = "vishwas"
>>> type(v1)
<class 'str'>
>>> v2 = 1234
>>> type(v2)
<class 'int'>
>>> v3 = 1234.5
>>> type(v3)
<class 'float'>
>>> v4 = True
>>> type(v4)
<class 'bool'>
>>> v5 = 23+6j
>>> type(v5)
<class 'complex'>
>>> |
```

```
<class 'complex'>
>>> v6 = 12345678908765432134567890987654321234567890987654321234567898765432457898765432
>>> v6
12345678908765432134567890987654321234567890987654321234567898765432457898765432
>>> v7 = 0b1010
>>> v7
10
>>> type(v7)
<class 'int'>
>>> v7 = 0x1a
>>> v7
26
>>> type(v7)
<class 'int'>
>>> v7 = 0o1234
>>> v7
668
>>> type(v7)
<class 'int'>
>>> v7 = 30_00_000
>>> v7
3000000
>>> type(v7)
<class 'int'>
```

Strings

```
>>> type(s1)
<class 'str'>
>>> s2 = 'vishwas'
>>> s2
'vishwas'
>>> type(s2)
<class 'str'>
>>> s3 = """Vishwas"""
>>> s3
'Vishwas'
>>> type(s3)
<class 'str'>
>>> s4 = '''Vishwas'''
>>> s4
'Vishwas'
>>> type(s4)
<class 'str'>
>>> s5 = "vishwas's book"
>>> s5
"vishwas's book"
>>> s6 = 'vishwas said "I am Vishwas, I am learning Python"'
>>> s6
'vishwas said "I am Vishwas, I am learning Python"'
>>> s7 = '''This is an example fo multiline string.
... This was created using triple quotes.
... This is a common scenario when text processing'''
>>> s7
'This is an example fo multiline string.\nThis was created using triple quotes.\nThis is a common scenario when text processing'
>>> |
```


Special Strings

```
>>> s10 = "C:\Users\VishwasKSingh\Workspace\ey-coh6-workspace\README.md"
File "<python-input-86>", line 1
    s10 = "C:\Users\VishwasKSingh\Workspace\ey-coh6-workspace\README.md"
          ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
SyntaxError: (unicode error) 'unicodeescape' codec can't decode bytes in position 2-3: truncated \UXXXXXXX escape
>>> s10 = r"C:\Users\VishwasKSingh\Workspace\ey-coh6-workspace\README.md"
>>> s10
'C:\\Users\\VishwasKSingh\\Workspace\\ey-coh6-workspace\\README.md'
>>> name = "viswhas"
>>> s11 = f"This is {name}'s book"
>>> s11
"This is viswhas's book"
>>> |
```

Operators

```
>>> 2+2
4
>>> 2-1
1
>>> 4*3
12
>>> 7/2
3.5
>>> 7//2
3
>>> 7%2
1
>>> 2**3
8
>>> |
```

```
>>> 2 > 3
False
>>> 3 > 4
False
>>> 3 < 5
True
>>> 3 >= 2
True
>>> 3 <= 5
True
>>> 3 != 5
True
>>> 4 == 4
True
>>> |
```

```
>>> 2 > 3 and 3 < 5
False
>>> 8 != 8 and 5 < 6
False
>>> |
```

- Non Zero value is always true
- 0 is always false

2 and 3	2 or 4	0 or 9
2 and 5	4 or 2	9 or 0
5 and 2	9 or 2	0 and 9
5 and 3	2 or 9	9 and 0

-1 and 5
5 and -1

Short circuit evaluation

- And Condition

expr1 and expr2

expr2 is evaluated only when expr1 is true

- Or Condition

expr1 or expr2

expr2 is not evaluated at all if expr1 is true

PEDMAS

23 / 6 * 4 + 3 ** (2 * 2)

Console I/O

- I can take a value from the user using console

- input(prompt) → is a function to do that

- it always returns a str

- you have to convert it explicitly

```
>>> print("Vishwas","k","singh")
Vishwas k singh
>>> print("Vishwas","k","singh", sep="*")
Vishwas*k*singh
>>> print("Vishwas","k","singh", sep="*", end="---")
>>> was*k*singh---
```

```
>>> x = 10
>>> y = 2
>>> x / y
5.0
>>> y = 0
>>> x / y
Traceback (most recent call last):
  File "<python-input-124>", line 1, in <module>
    x/y
    ~^~
ZeroDivisionError: division by zero
>>> y != 0 and x / y
False
>>> y = 2
>>> y != 0 and x / y
5.0
>>> |
```

Simple Interest Formula: $si = (p \cdot t \cdot r) / 100$

SIP vs Lumpsum calculator

Users	City	
Vishwas	Pune	
Raju	Goa	
Sachin	Delhi	